CHEM 1211K Exam 2 — Concept-Based Practice Set

Each block starts with the original question number, then the **concept** that question tests, then a short note on how the new problems come at it **from a different angle**. Answers are in the separate answer-key PDF.

Constants: R = 0.08206 L·atm/(mol·K) = 8.314 J/(mol·K); h = 6.626×10^{-34} J·s; c = 2.998×10^{8} m/s; N_A = 6.022×10^{23} mol^{-1}; 1 eV = 1.602×10^{-19} J; 1 L·atm = 101.3 J; K = °C + 273.15; molar volume at STP = 22.41 L/mol.

Part I — Solutions and Stoichiometry

Original Question 1 · Dilution

Concept tested: Moles of solute do not change when you add solvent, so $M_1V_1 = M_2V_2$ (any volume units work, as long as V_1 and V_2 match).
New angle: The original asks for the stock volume. These ask for the final concentration, use water added rather than final volume, and chain two dilutions.

1A. 25.0 mL of 12.0 M HCl is diluted to a final volume of 0.500 L. What is the molarity of the diluted acid?

1B. A student adds 150.0 mL of water to 50.0 mL of 1.80 M KNO_3. Assuming volumes are additive, what is the new concentration? (Watch out: 150.0 mL is not the final volume.)

1C. Serial dilution: 10.0 mL of 1.00 M $CuSO_4$ is diluted to 100.0 mL (solution A). Then 5.00 mL of solution A is diluted to 250.0 mL (solution B). What is the molarity of solution B?

Original Question 2 · Gas stoichiometry with a limiting reactant

Concept tested: Converting every reactant to moles by the right route (PV = nRT for gases, M × V for solutions, mass/molar mass for solids), then using mole ratios to find the limiting reactant and the yield.
New angle: Here you work backward (find the gas volume needed), and a second problem asks for a gas product volume plus the leftover excess reactant.

2A. $CO_2(g) + 2\ NaOH(aq) \rightarrow Na_2CO_3(aq) + H_2O(l)$
What volume (L) of CO_2 at 25.0 °C and 0.980 atm is needed to react completely with 0.250 L of 1.20 M NaOH?

2B. $Zn(s) + 2\ HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$
3.27 g of Zn is added to 50.0 mL of 1.50 M HCl. (a) Which reactant is limiting? (b) What volume of H_2 forms at STP (0 °C, 1.00 atm)? (c) How many moles of the excess reactant remain?

Original Question 3 · Ion concentrations after mixing (common ion)

Concept tested: Each compound dissociates into a set number of ions. When two solutions share an ion, you add the moles of that ion from each source, then divide by the total final volume.
New angle: Same idea with K+ as the shared ion, then the reverse: how much solid must you add to reach a target ion concentration?

3A. 40.0 mL of 0.500 M K_2SO_4 is mixed with 60.0 mL of 0.300 M KCl (total volume 100.0 mL). Find $[K^+]$, $[SO_4^{2-}]$, and $[Cl^-]$.

3B. How many grams of $MgCl_2$ (95.21 g/mol) must be dissolved in 250.0 mL of 0.100 M NaCl so that the final $[Cl^-]$ is 0.500 M? Assume the volume does not change.

Original Question 4 · Molarity and ions from a formula

Concept tested: Molarity = mol/L, and the subscripts in a formula tell you how many of each ion one formula unit releases.
New angle: Start from an ion concentration and work back to the compound, then compare total ion concentrations across different salts.

4A. A solution of $Fe_2(SO_4)_3$ has $[SO_4^{2-}] = 0.450$ M. (a) What is $[Fe_2(SO_4)_3]$? (b) What is $[Fe^{3+}]$? (c) What mass of $Fe_2(SO_4)_3$ (399.88 g/mol) is in 500.0 mL of this solution?

4B. Which solution has the highest total ion concentration? 0.10 M Na_3PO_4, 0.15 M $CaCl_2$, 0.20 M KNO_3, or 0.10 M $Al_2(SO_4)_3$.

Part II — Thermochemistry

Original Question 5 · Open, closed, and isolated systems

Concept tested: Open: matter and energy can cross the boundary. Closed: energy only. Isolated: neither.
New angle: Instead of classifying, you reason about what crosses the boundary and change the setup.

5A. Classify each: (a) a sealed can of soda cooling in a refrigerator (b) a human body (c) a candle burning inside a sealed glass jar (d) a perfectly insulated, sealed thermos (e) a pot of soup with the lid off on a stove.

5B. A pot of water boils on a stove with its lid off. What single change would turn it into a closed system? What would make it (ideally) isolated?

Original Question 6 · Signs of q, w, and ΔU

Concept tested: $\Delta U = q + w$ with the chemist's sign convention: $q > 0$ when the system absorbs heat, $w > 0$ when work is done on the system. At constant volume, $w = 0$ so $\Delta U = q$.
New angle: The original was constant volume ($w = 0$). These involve expansion and compression, so w is not zero and you must calculate it with $w = -P\Delta V$.

6A. A gas in a piston absorbs 450 J of heat while expanding from 2.00 L to 5.50 L against a constant external pressure of 1.20 atm. Find q, w (in J), and ΔU. (1 L·atm = 101.3 J)

6B. The surroundings compress a gas, doing 325 J of work on it, while the gas releases 510 J of heat. Give the signs and values of q, w, and ΔU.

6C. An exothermic reaction happens in a sealed, rigid steel container. What are the signs of q, w, and ΔU?

Original Question 7 · Hess's law

Concept tested: If you reverse an equation, flip the sign of ΔH. If you multiply an equation, multiply ΔH by the same factor. Then add the steps so unwanted species cancel.
New angle: The first problem finds the ΔH of a step you cannot measure directly; the second requires both reversing and scaling.

7A. Use these equations to find ΔH for $C(s) + \frac{1}{2}\ O_2(g) \rightarrow CO(g)$:
(1) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H = -393.5$ kJ
(2) $CO(g) + \frac{1}{2}\ O_2(g) \rightarrow CO_2(g)$ $\Delta H = -283.0$ kJ

7B. Given:
(1) $N_2(g) + O_2(g) \rightarrow 2\ NO(g)$ $\Delta H = +180.6$ kJ
(2) $2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$ $\Delta H = -114.2$ kJ
Find ΔH for $NO_2(g) \rightarrow \frac{1}{2}\ N_2(g) + O_2(g)$.

Original Question 8 · Coffee-cup calorimetry (neutralization)

Concept tested: Heat released by the reaction = −(heat absorbed by the solution + heat absorbed by the calorimeter): $q_{rxn} = -(mc\Delta T + C_{cal}\Delta T)$. Divide by moles of the limiting reactant (or water formed) to get ΔH in kJ/mol.
New angle: Run the problem forward (predict the final temperature from ΔH) and solve for the calorimeter's heat capacity.

8A. 50.0 mL of 1.00 M HCl and 75.0 mL of 0.800 M NaOH, both at 22.15 °C, are mixed in a calorimeter with $C_{cal} = 18.0$ J/°C. $\Delta H_{neut} = -55.8$ kJ per mol H_2O. Density = 1.00 g/mL, c = 4.184 J/(g·°C). What is the final temperature?

8B. To find a calorimeter's heat capacity, 50.0 g of water at 60.0 °C is poured into 50.0 g of water at 20.0 °C already inside the calorimeter (also at 20.0 °C). The final temperature is 39.2 °C. What is C_{cal}?

Original Question 9 · ΔH°rxn from enthalpies of formation

Concept tested: $\Delta H°rxn = \Sigma\ n\ \Delta H°_f(\text{products}) - \Sigma\ n\ \Delta H°_f(\text{reactants})$. Elements in their standard state have $\Delta H°_f = 0$.
New angle: Solve for an unknown $\Delta H°_f$ when $\Delta H°rxn$ is given, and identify which substances have $\Delta H°_f = 0$.

9A. $C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(l)$ $\Delta H°rxn = -1366.8$ kJ
Given $\Delta H°_f[CO_2(g)] = -393.5$ kJ/mol and $\Delta H°_f[H_2O(l)] = -285.8$ kJ/mol, find $\Delta H°_f$ of ethanol.

9B. Which of these have $\Delta H°_f = 0$? $O_2(g)$, $O_3(g)$, C(graphite), C(diamond), $Br_2(l)$, $Br_2(g)$, Na(s), H(g)

Original Question 10 · Specific heat: q = mcΔT

Concept tested: q = mcΔT links heat, mass, specific heat, and temperature change. If you know any three, you can find the fourth. Heat lost by one object = heat gained by another.
New angle: Solve for c using heat transfer between a metal and water, find q for cooling, and compare two metals.

10A. A 25.0 g metal sample at 98.0 °C is dropped into 100.0 g of water at 22.0 °C. The final temperature is 24.3 °C. What is the specific heat of the metal?

10B. How much heat (kJ) must be removed to cool 250.0 g of water from 85.0 °C to 25.0 °C?

10C. The same amount of heat is added to 10.0 g of iron (c = 0.450 J/g·°C) and 10.0 g of aluminum (c = 0.897 J/g·°C). Which warms up more, and by roughly what factor?

Original Question 11 · Thermochemical stoichiometry

Concept tested: A thermochemical equation's ΔH is the heat for the mole amounts written in it. Convert to moles, then use ΔH as a conversion factor (kJ per mol of reaction).
New angle: Find the mass needed to give a set amount of heat, start from a gas volume, and deal with a limiting reactant.

11A. $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$ $\Delta H = -890.4$ kJ
What mass of CH_4 (16.04 g/mol) must burn to release 5.00×10^3 kJ?

11B. Using $\Delta H = -2220$ kJ for C_3H_8 combustion (same equation as the original question), how much heat is released when 15.0 L of propane at 25.0 °C and 1.00 atm is burned?

11C. $2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(l)$ $\Delta H = -571.6$ kJ
How much heat is released when 4.00 g of H_2 reacts with 16.0 g of O_2?

Part III — Gases

Original Question 12 · Kinetic molecular theory postulates

Concept tested: Knowing the exact postulates: random motion, negligible molecular volume, no attractive or repulsive forces, perfectly elastic collisions, and KE_{avg} directly proportional to Kelvin temperature. Test makers swap words like elastic/inelastic or direct/inverse.
New angle: Instead of picking the true statement, fix the false ones and link each postulate to a real observation.

12A. Rewrite each false statement so it is true:

(a) Gas molecules move in uniform, predictable paths.
(b) Collisions between gas molecules are perfectly inelastic.
(c) Average kinetic energy is inversely proportional to Kelvin temperature.
(d) Gas molecules attract each other strongly during collisions.

12B. Which postulate best explains each observation? (a) Gases are easy to compress. (b) A sealed gas at constant temperature keeps the same pressure forever instead of "running down." (c) Doubling the Kelvin temperature doubles the average kinetic energy.

Original Question 13 · Real vs. ideal gases

Concept tested: Real gases deviate when molecules are close together (high P) and moving slowly enough for attractions to matter (low T). Larger, more polar molecules deviate more.
New angle: Rank conditions, compare gases, and explain what each van der Waals constant corrects for.

13A. Rank these from most ideal to least ideal behavior for N_2: (A) 500 K, 0.5 atm (B) 200 K, 50 atm (C) 300 K, 1 atm (D) 500 K, 50 atm

13B. At the same T and P, which deviates more from ideal behavior: He or SO_2? Explain using a and b.

13C. In the van der Waals equation, what does each constant correct for? Does the a-term make the real pressure higher or lower than ideal?

Original Question 14 · Van der Waals calculation

Concept tested: $P = nRT/(V - nb) - a(n/V)^2$. You must plug in carefully and keep units consistent (L, atm, mol, K).
New angle: Compare the real and ideal answers to see how big the correction is and which term wins.

14A. Calculate the pressure of 2.00 mol of CO_2 in 1.50 L at 300.0 K (a) with the ideal gas law and (b) with the van der Waals equation ($a = 3.592$ atm·L²/mol², $b = 0.04267$ L/mol). (c) What is the percent difference, and which correction dominates?

Original Question 15 · KE_{avg} depends on T; speed depends on T and molar mass

Concept tested: At the same T, all gases have the same KE_{avg}. Lighter gases move faster ($v_{rms} = \sqrt{(3RT/M)}$). Pressure depends on n, T, and V, not on the identity of the gas.
New angle: Now the gas is the same but n and T change, then calculate v_{rms} and find a matching temperature.

15A. Container A holds 2 mol N_2 at 300 K; container B (same volume) holds 1 mol N_2 at 600 K. Compare (a) P (b) KE_{avg} (c) v_{rms}.

15B. Calculate v_{rms} of O_2 at 25.0 °C ($R = 8.314$ J/mol·K, M in kg/mol). At what temperature would He (4.003 g/mol) have the same v_{rms}?

Original Question 16 · Charles's law and combined gas law

Concept tested: At constant P and n, V/T is constant, and T must be in Kelvin. The combined law $P_1V_1/T_1 = P_2V_2/T_2$ extends it.
New angle: Solve for a volume, use the combined law when two variables change, and see why °C fails.

16A. A balloon has a volume of 2.50 L at 22.0 °C. It is taken outside where it is −15.0 °C (P constant). What is its new volume?

16B. A weather balloon holds 125 L of He at 1.00 atm and 20.0 °C. It rises to where P = 0.350 atm and T = −40.0 °C. What is its volume?

16C. A student says “heating a gas from 20 °C to 40 °C doubles its volume.” Is that right? By what factor does V actually change?

Original Question 17 · Dalton's law and mole fractions

Concept tested: $P_i = \chi_i P_{total}$, so partial pressure is proportional to moles. Equal moles give equal partial pressures.
New angle: Start from masses, collect a gas over water, and compare equal masses instead of equal moles.

17A. 6.00 g of Ne and 8.00 g of O_2 are placed in a 5.00 L flask at 25.0 °C. Find the total pressure and each partial pressure.

17B. H_2 is collected over water at 25.0 °C. The total pressure is 755.0 mmHg and the vapor pressure of water is 23.8 mmHg. If 0.250 L of gas is collected, how many moles of H_2 are there?

17C. A flask holds equal masses of He and Ar. Which has the larger partial pressure, and by about how much?

Original Question 18 · Graham's law (find a molar mass)

Concept tested: $rate_1/rate_2 = \sqrt{(M_2/M_1)}$. Lighter gases effuse faster. Time is the inverse of rate.
New angle: Use effusion time instead of rate, identify a hydrocarbon formula, and compare isotopes.

18A. An unknown noble gas takes 2.24 times as long to effuse as He under the same conditions. Identify it.

18B. An alkane C_nH_{2n+2} effuses at 0.742 times the rate of O_2. Find its molar mass and formula.

18C. Uranium enrichment separates $^{235}UF_6$ (349.03 g/mol) from $^{238}UF_6$ (352.04 g/mol) by effusion. What is the ratio of their effusion rates?

Original Question 19 · Effusion rate depends on both T and M

Concept tested: rate ∝ √(T/M). A heavier gas can still effuse quickly if it is hot enough, so you must compare T/M, not just M.
New angle: Find gases with equal rates, and solve for the temperature that makes two gases match.

19A. Which two pairs have essentially equal effusion rates? H_2 at 100 K, He at 200 K, CH_4 at 400 K, O_2 at 800 K

19B. At what temperature would SO_2 (64.07 g/mol) effuse at the same rate as CH_4 (16.04 g/mol) at 300 K?

Part IV — Entropy and Gibbs Free Energy

Original Question 20 · Predicting the sign of ΔS

Concept tested: Entropy increases with more gas moles, solid → liquid → gas, dissolving, higher temperature, and more complex molecules.
New angle: Includes cases without a phase change (temperature change, gas moles decreasing) and asks you to rank absolute entropies.

20A. Predict the sign of ΔS: (a) $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ (b) NaCl(s) dissolving in water (c) $H_2O(l)$ heated from 25 °C to 75 °C (d) $2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$ (e) frost forming on a car window

20B. Rank by standard molar entropy, lowest to highest: (a) $H_2O(g)$, $H_2O(s)$, $H_2O(l)$ (b) $C_3H_8(g)$, $CH_4(g)$, $C_2H_6(g)$

Original Question 21 · ΔG = ΔH − TΔS and the crossover temperature

Concept tested: ΔG° is linear in T with slope −ΔS and intercept ΔH. At the crossover temperature ΔG = 0, so T = ΔH/ΔS. (In the original: ΔH = 100ΔS and 65.0 = ΔH − 293ΔS give ΔS ≈ −337 J/mol·K.)
New angle: Given ΔG at two temperatures, find ΔH and ΔS. Then do a simple crossover calculation.

21A. A reaction has ΔG° = −12.0 kJ/mol at 300 K and +8.0 kJ/mol at 500 K. Assuming ΔH° and ΔS° are constant, find ΔS°, ΔH°, and the crossover temperature. Is it spontaneous above or below that temperature?

21B. $\Delta H^\circ = +45.0$ kJ/mol and $\Delta S^\circ = +125$ J/(mol·K). At what temperature does the reaction become spontaneous, and in which direction of temperature change?

Original Question 22 · Phase transitions: T = ΔH/ΔS

Concept tested: At the normal boiling or melting point the two phases are in equilibrium, so $\Delta G = 0$ and $T = \Delta H/\Delta S$ (convert J to kJ!).
New angle: Solve for ΔS from a known boiling point, then decide which direction a phase change goes away from equilibrium.

22A. Ethanol boils at 78.4 °C with $\Delta H_{vap} = 38.6$ kJ/mol. What is ΔS_{vap}?

22B. For ice: $\Delta H_{fus} = 6.01$ kJ/mol, $\Delta S_{fus} = 22.0$ J/(mol·K). (a) Calculate the melting point. (b) Calculate ΔG for melting at −10.0 °C. Is melting or freezing spontaneous there?

Original Question 23 · Temperature dependence of spontaneity

Concept tested: With ΔH and ΔS having the same sign, the temperature decides. In a plot of ΔG vs T, the slope is $-\Delta S$ and the y-intercept is ΔH.
New angle: Use a reaction with both signs negative (the opposite of $CaCO_3$), and read signs off a graph description.

23A. $2\ NO_2(g) \rightarrow N_2O_4(g)$ $\Delta H^\circ = -57.2$ kJ, $\Delta S^\circ = -175.8$ J/K
(a) Is it spontaneous at 25 °C? (b) At 100 °C? (c) What is the crossover T? (d) Is the slope of ΔG vs T positive or negative?

23B. A plot of ΔG° vs T is a straight line with a negative slope and a y-intercept of +50 kJ. What are the signs of ΔH and ΔS? Is the reaction favored at high or low T?

Original Question 24 · The four ΔH/ΔS sign combinations

Concept tested: (−,+): always spontaneous. (+,−): never. (−,−): low T only. (+,+): high T only.
New angle: Fill in the other two combinations and apply them to real reactions.

24A. Reaction C is exothermic with $\Delta S < 0$. Reaction D is endothermic with $\Delta S < 0$. Which is never spontaneous? When is the other one spontaneous?

24B. Classify: (a) $2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$, $\Delta H < 0$ (b) photosynthesis, $6\ CO_2(g) + 6\ H_2O(l) \rightarrow C_6H_{12}O_6(s) + 6\ O_2(g)$, $\Delta H > 0$

Original Question 25 · Rearranging ΔG = ΔH − TΔS

Concept tested: Solve the equation for any variable and keep the units consistent (J vs kJ, K not °C).
New angle: Solve for ΔH and for T instead of ΔS.

25A. For $N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$ at 25.00 °C, $\Delta G° = -33.0$ kJ and $\Delta S° = -198.7$ J/K. Find $\Delta H°$.

25B. A reaction has $\Delta H° = -120.0$ kJ and $\Delta S° = -150.0$ J/K. At what temperature is $\Delta G° = -60.0$ kJ?

Original Question 26 · Full thermodynamic analysis from tables

Concept tested: Use the same products-minus-reactants pattern for both $\Delta H°_f$ and S°, remembering that S° of elements is not zero. Then use $\Delta G = \Delta H - T\Delta S$.
New angle: A new reaction where you also find the temperature at which spontaneity flips, and a problem that back-solves a missing S°.

26A. $2\ SO_2(g) + O_2(g) \rightarrow 2\ SO_3(g)$
Data: SO_2: $\Delta H°_f = -296.8$ kJ/mol, S° = 248.2 J/(mol·K); O_2: S° = 205.0; SO_3: $\Delta H°_f = -395.7$, S° = 256.8
(a) $\Delta H°$ (b) $\Delta S°$ (c) $\Delta G°$ at 298 K, and is it spontaneous? (d) Above what temperature does it become nonspontaneous?

26B. For $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$, $\Delta S° = -242.8$ J/K. Given S°: $CO_2 = 213.7$, $H_2O(l) = 69.9$, $O_2 = 205.0$ J/(mol·K), find S° of $CH_4(g)$.

Part V — Light and Photons

Original Question 27 · Visible spectrum: wavelength, frequency, energy

Concept tested: λ and ν are inversely related ($c = \lambda\nu$), and $E = h\nu$. In visible light, red has the longest λ and lowest energy, and violet has the shortest λ and highest energy (ROYGBIV).
New angle: Ask for the opposite end of the spectrum, use numbers, and extend to the whole EM spectrum.

27A. Which visible color has (a) the shortest wavelength (b) the lowest frequency (c) the lowest energy per photon?

27B. Light A is 650 nm and light B is 450 nm. Which is roughly red and which is blue? Which has the higher frequency?

27C. Rank by increasing photon energy: X-rays, microwaves, visible, radio, ultraviolet, gamma rays, infrared.

Original Question 28 · What controls photon energy

Concept tested: $E = h\nu = hc/\lambda$. Only frequency (or wavelength) changes the energy per photon. Amplitude/intensity changes the number of photons, not the energy of each one, and h is a constant.
New angle: Flip it to increasing energy and separate brightness from color.

28A. Select all that INCREASE the energy of a single photon: (a) increase wavelength (b) increase frequency (c) increase intensity (d) decrease wavelength (e) shift from red light to blue light

28B. A dim violet light and a very bright red light shine on a surface. Which has more energetic photons? Which delivers more photons per second?

Original Question 29 · Energy per mole of photons

Concept tested: E (one photon) = hν or hc/λ. Multiply by Avogadro's number for a mole of photons, and convert units ($1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$).
New angle: Start from wavelength instead of frequency, convert eV to wavelength, and count photons.

29A. What is the energy in kJ/mol of photons with $\lambda = 532$ nm (green laser)?

29B. A photon has an energy of 2.50 eV. What is its wavelength in nm?

29C. How many photons are in a 1.00 mJ pulse of 405 nm laser light?

Original Question 30 · Wavelength from energy: λ = hc/E

Concept tested: Rearranging E = hc/λ and converting m to nm (1 nm = 10^{-9} m).
New angle: Use a bond energy per mole (convert to per molecule first) and a frequency in a different region of the spectrum.

30A. The C–C bond energy is 348 kJ/mol. What is the longest wavelength (nm) of light that can break one C–C bond? What region of the spectrum is this?

30B. An FM radio station broadcasts at 101.1 MHz. Find the wavelength (m) and the energy of one photon (J).